

Correction - Distributivité simple

Exercice 1

$$\begin{aligned}\text{a. } 3(x + 4) \\ &= 3 \times x + 3 \times 4 \\ &= 3x + 12\end{aligned}$$

$$\begin{aligned}\text{b. } 5(y - 2) \\ &= 5 \times y - 5 \times 2 \\ &= 5y - 10\end{aligned}$$

$$\begin{aligned}\text{c. } 7(2x + 1) \\ &= 7 \times 2x + 7 \times 1 \\ &= 14x + 7\end{aligned}$$

$$\begin{aligned}\text{d. } -2(x + 6) \\ &= -2 \times x + (-2) \times 6 \\ &= -2x - 12\end{aligned}$$

$$\begin{aligned}\text{e. } -4(3 - a) \\ &= -4 \times 3 - (-4) \times a \\ &= -12 + 4a\end{aligned}$$

$$\begin{aligned}\text{f. } x(x + 5) \\ &= x \times x + x \times 5 \\ &= x^2 + 5x\end{aligned}$$

Exercice 2

$$\begin{aligned}A &= 2(x + 3) + 5x \\ A &= 2 \times x + 2 \times 3 + 5x \\ A &= 2x + 6 + 5x \\ A &= 7x + 6\end{aligned}$$

$$\begin{aligned}B &= 4(2x - 1) - 3 \\ B &= 4 \times 2x - 4 \times 1 - 3 \\ B &= 8x - 4 - 3 \\ B &= 8x - 7\end{aligned}$$

$$\begin{aligned}C &= 3(x + 2) + 2(x - 5) \\ C &= 3 \times x + 3 \times 2 + 2 \times x - 2 \times 5 \\ C &= 3x + 6 + 2x - 10 \\ C &= 5x - 4\end{aligned}$$

$$\begin{aligned}D &= 5(2y + 1) - 3(y + 4) \\ D &= 5 \times 2y + 5 \times 1 - 3 \times y - 3 \times 4 \\ D &= 10y + 5 - 3y - 12 \\ D &= 7y - 7\end{aligned}$$

$$\begin{aligned}E &= -2(3x - 4) + 6x \\ E &= -2 \times 3x - (-2) \times 4 + 6x \\ E &= -6x + 8 + 6x \\ E &= 8\end{aligned}$$

$$\begin{aligned}F &= x(x + 2) + 3(x - 1) \\ F &= x \times x + x \times 2 + 3 \times x - 3 \times 1 \\ F &= x^2 + 2x + 3x - 3 \\ F &= x^2 + 5x - 3\end{aligned}$$

Exercice 3

$$\begin{aligned}G &= 5 + (3x - 2) \\ G &= 5 + 3x - 2 \\ G &= 3x + 3\end{aligned}$$

$$\begin{aligned}H &= 7x - (2x + 4) \\ H &= 7x - 2x - 4 \\ H &= 5x - 4\end{aligned}$$

$$\begin{aligned}I &= 4 - (x - 6) + 3x \\ I &= 4 - x + 6 + 3x \\ I &= 2x + 10\end{aligned}$$

Exercice 4

1. Aire = $3(x + 5)$

2. $3(x + 5)$

= $3 \times x + 3 \times 5$

= $3x + 15$

3. Pour $x = 2$: $3 \times 2 + 15 = 6 + 15 = 21$

L'aire du rectangle est 21 cm^2 .